

Lab 4: Simulation in Quantum Mechanics

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1

We simulated the interference of a Gaussian wave packet colliding with an infinite potential wall using different number of mesh points $N \in \{2^{13}, 2^{14}\}$. The figures below show the interference patterns at $t = 95$.

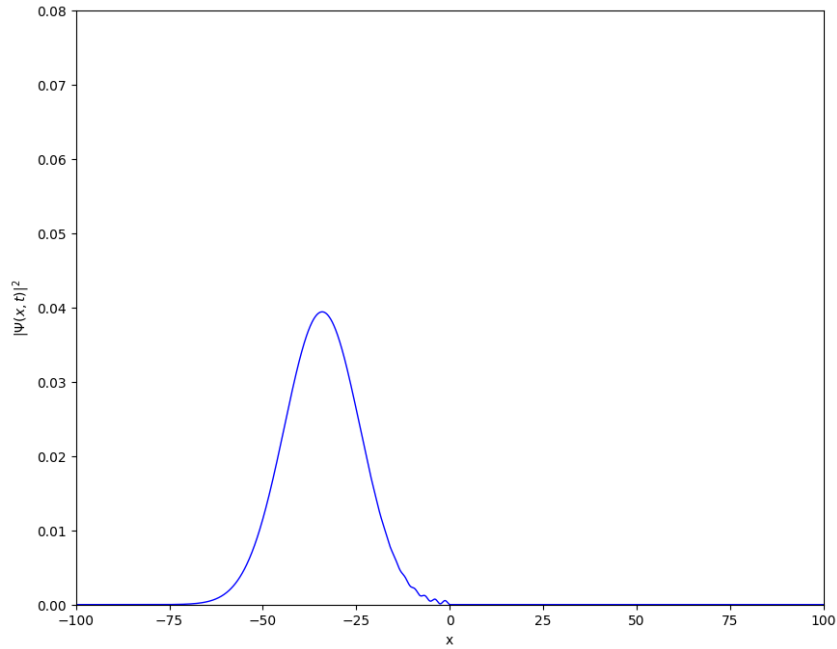


Figure 1: Interference pattern for $N = 2^{13}$

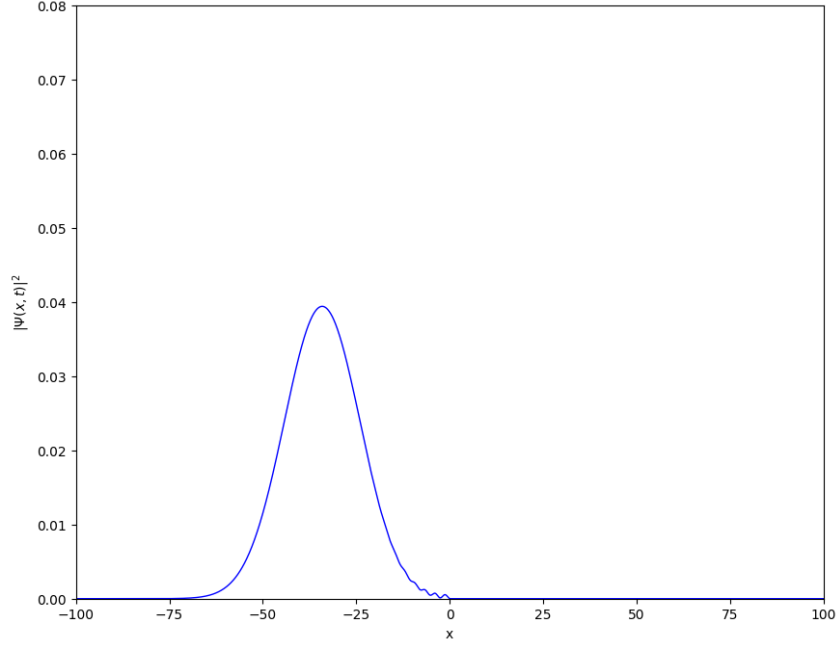


Figure 2: Interference pattern for $N = 2^{14}$

The figures for $N = 2^{13}$ and $N = 2^{14}$ are nearly identical. This suggests that the interference results are physical and not a consequence of discretization or numerical parameters.

2

We simulated the wave packet's interaction with the Coulomb barrier potential, defined as $V(x) = 0$ for $x < 1$ and $V(x) = 1/x$ for $x \geq 1$. Figure 3 shows the probability density as the wave packet split into reflected and transmitted parts.

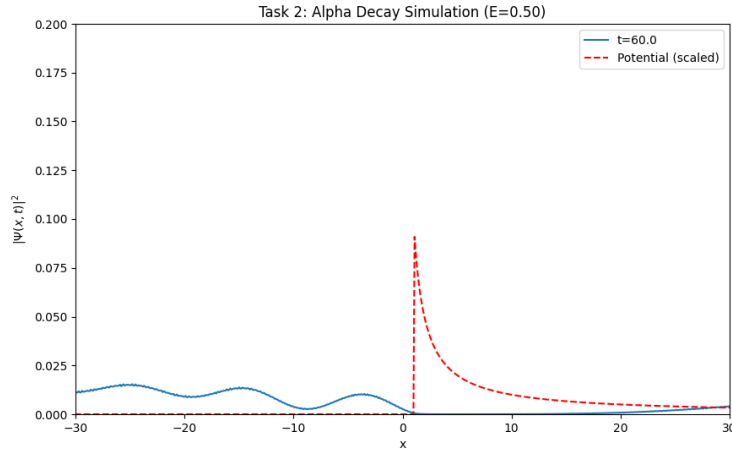


Figure 3: Wave packet splitting at the Coulomb barrier ($E = 0.5$)

To calculate the transmission probability T , we use the relation:

$$T = \frac{Y_T}{Y_R + Y_T} \quad (1)$$

where Y_R and Y_T are the integrated probability densities (or maximum amplitudes in a steady state, but here we use the integrated area) for the reflected and transmitted parts respectively. According to Gamow's theory, T should vary exponentially with energy E :

$$T \propto \exp\left(-\frac{k}{\sqrt{E}}\right) \quad (2)$$

We plotted $\ln T$ against $1/\sqrt{E}$ in Figure 4.

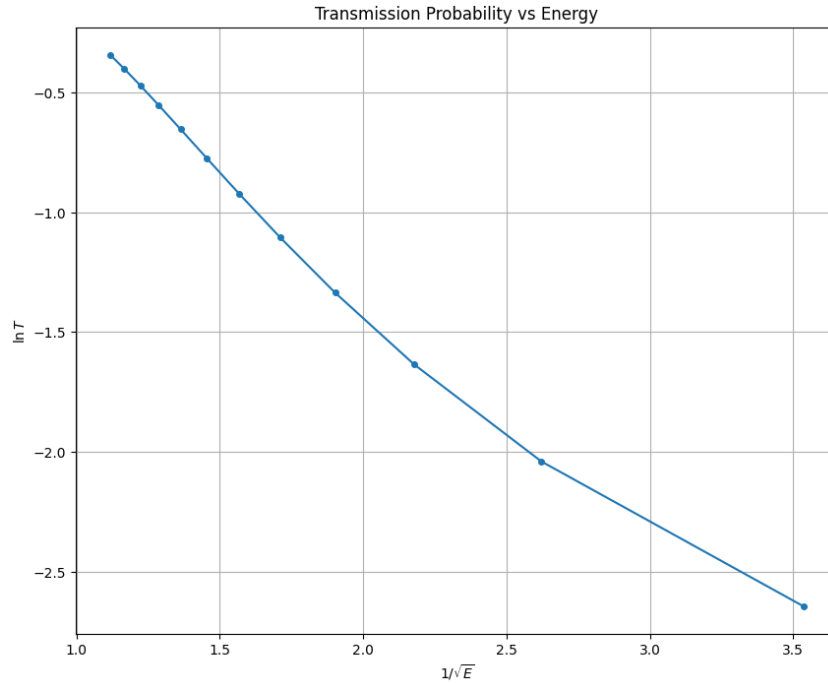


Figure 4: Transmission probability vs $1/\sqrt{E}$

The plot shows a linear relationship for the investigated range, agreeing with Gamow's theoretical prediction.

3

We initialized a Gaussian wave packet with energy $E = 0.5$ (incident wave vector $k_0 = 1$). The potential step was defined as $V(x) = 0$ for $x < 0$ and $V(x) = 1$ for $x \geq 0$.

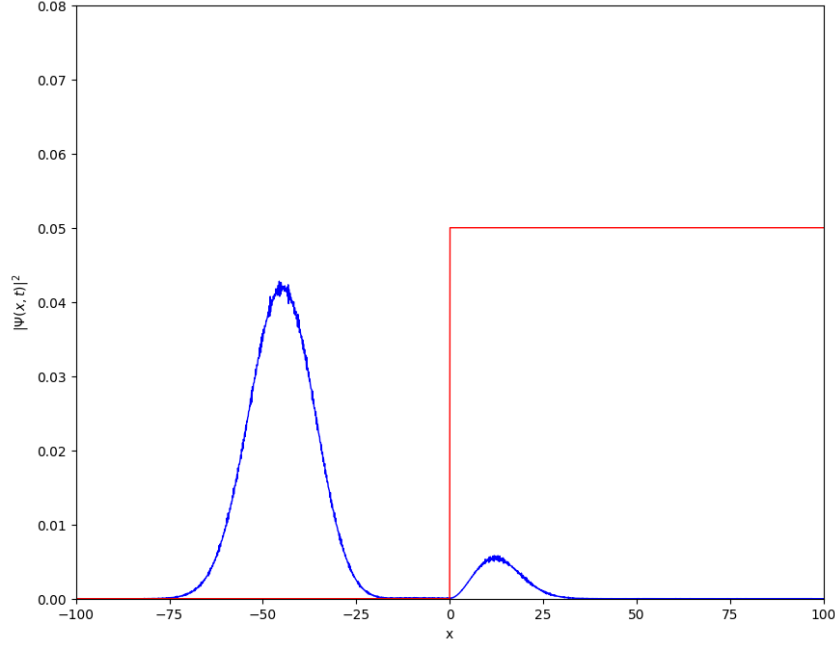


Figure 5: Reflection from a potential step ($E < V_0$)

As seen in Figure 5, since the energy $E = 0.5$ is less than the barrier height $V_0 = 1$, the wave packet is almost entirely reflected.

Next, we applied an electric field along the x-direction, resulting in a triangular potential barrier $V(x) = V_0(1 - x/L)$ for $0 < x < L$ and $V(x) = 0$ otherwise, with $L = 4$. We plotted $\ln T$ against $V_0^{3/2}$ for $V_0 \in \{0.6, 0.7, \dots, 1.2\}$.

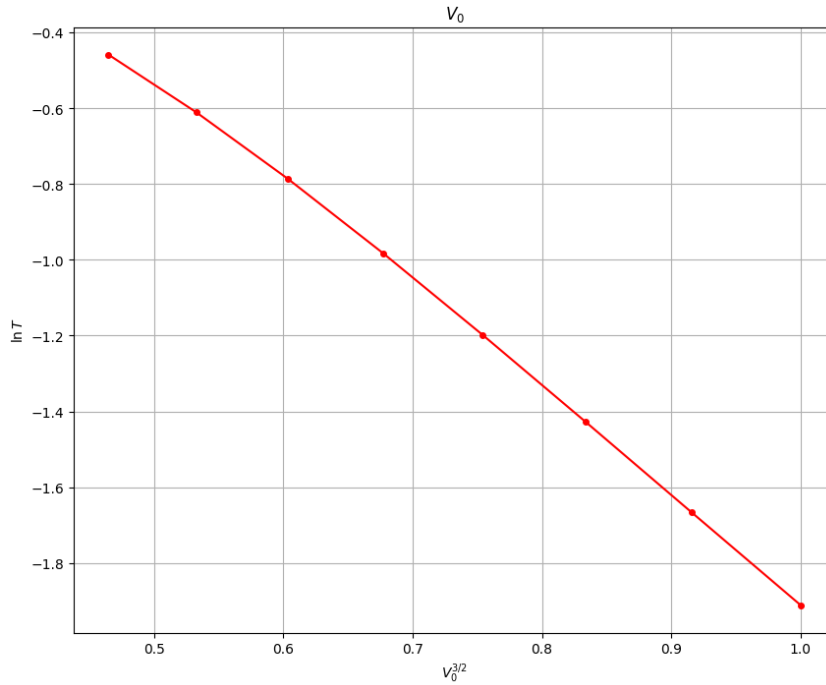


Figure 6: $\ln T$ vs $V_0^{3/2}$ for Fowler-Nordheim tunneling

The linear relationship in Figure 6 agrees with the WKB approximation for field emission, where the transmission probability decreases exponentially with $V_0^{3/2}$.